

Understanding the v12 SageMath Verification Package

An explanatory guide to exact algebra, rigorous scalar intervals, reproducibility, and the remaining mathematical dependencies

80 PASS

Passing check groups

48 EXACT

Replayable certificates

7 HUMAN

Unformalized dependency groups

Actually executed in SageMath 10.9. Zero failed check groups.

This is not complete machine certification of the all-28 existence theorem. Computational evidence, scoped mathematical review and external dependencies are separated throughout.

SOURCE SHA-256

ea27de068cd78871211d0a1ac102e76b65c7bd1f033267fb24a2c5f8dc423f9e

Submitted source preserved byte-for-byte. Release date: 2026-09-25.

Reading guide

Read the mathematical chain in order, then use the appendices to locate every check and source formula.

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Appendix A: every check group. Appendix B: source formula inventory. Full evidence is in the accompanying JSON files.

CHAPTER 01

How to read the verification verdict

This report explains the computational companion to version 12 of the manuscript, rather than replacing the manuscript with a computer proof. The recorded run contains 80 passing check groups, 7 explicitly unverified dependency groups, and 48 independently replayable exact certificates. These are different counting units. A single passing group can contain thousands of component comparisons; a certificate can encode just one identity. Neither number measures the percentage of the main theorem proved.

The most useful interpretation is evidential. The package reconstructs local curvature expressions from geometric inputs, checks the algebra used to estimate arbitrary planes, verifies boundary cancellations, and tests selected scalar inequalities with rigorous intervals. It also records the exact source file, the executable environment, the outputs, and the independent replay procedure. A reader can therefore determine what was calculated and reproduce that calculation.

The important qualification is equally explicit. Smooth constructions, global quotient identifications, compactness arguments, analytic smoothing, and cited classification or gluing theorems are not made true by a successful script. They remain mathematical premises or arguments that must be read. UNVERIFIED means not proved by the computational component; it does not mean a counterexample was found. Conversely, PASS means the stated check succeeded, not that every possible error in the entire paper was excluded.

PASS $\neq$ complete formal proof

evidence = source + code + outputs + scope

- Read the claim and evidence attached to each result, not its status alone.
- Use the report together with the source manuscript and the dependency register.

Source and evidence

reproduction/reference-run/summary.json
 VERIFICATION_REPORT.md
 docs/PROOF_SCOPE.md

CHAPTER 02

Which manuscript and which changes?

Every verification statement in this release is attached to the supplied `v12_clean.tex`, preserved as `paper/submitted.tex`. Its SHA-256 digest is `ea27de068cd78871211d0a1ac102e76b65c7bd1f033267fb24a2c5f8dc423f9e`. The digest identifies a byte sequence: it prevents accidentally reporting an older paper's successful run as evidence for this one. It is not a mathematical correctness test. A different source, including one with an innocuous spacing edit, requires a new source identity record.

The source-binding module also compares the inventoried labeled displays with a separately frozen review contract. That contract makes changes to the selected formulas visible. Its coverage is deliberately described: inline formulas and unlabeled displays are not individually inventoried, although the full-source hash still binds them. Human comparison remains necessary to establish that a program expresses the correct mathematical object and not a similarly named formula.

For this v12 release, the supplied source and the copy named `revised.tex` are byte-for-byte identical. No substantive correction was justified by the scoped audits, and the package does not manufacture an editorial revision merely to appear productive. Earlier-release changes must not be imported silently. In particular, the boundary symbols now read `mu_N` and `mu_S`; the code's corresponding variables have been updated. The mathematical meanings are the angular logarithmic derivatives on the two sides.

SHA256(submitted. tex) = reviewed digest

$$\mu_N = \frac{F'(\ell_N)}{F_a}, \quad \mu_S = \cot a$$

- A source hash establishes identity, not truth.
- If a formula changes, review its translation and rerun the affected dependency chain.

Source and evidence

`checks/source_binding.py`
`docs/reviewed_formula_contract.json`
`paper/submitted.tex`

CHAPTER 03

The quantifiers in the all-types claim

The manuscript's target is existence of a smooth metric with strictly positive sectional curvature on every oriented smooth homotopy seven-sphere type. The finite label group has 28 elements. Reversing orientation does not change whether a metric has positive sectional curvature, but that observation does not remove the obligation to identify the geometric models used in the proof. Counting 28 residues alone cannot establish the theorem.

The construction is parameterized by an integer n and uses an attaching map identified with the n th iterate of the Durán-Püttmann-Rigas generator. The topology argument, together with that external classification statement and the chosen orientation convention, supplies the passage from n to a smooth type modulo 28. The computational enumeration checks the bookkeeping once this correspondence is granted; it does not independently classify exotic spheres.

For each fixed n , all geometric constants may depend on the chosen connection and therefore on n . The proof does not need one numerical epsilon that works for every representation or every integer simultaneously. It does need one positive epsilon for the chosen closed manifold that works at every point and every two-plane. This distinction is central: pointwise success with a plane-dependent parameter would not establish strictly positive curvature of a single metric.

$$\forall n \in \mathbb{Z}/28\mathbb{Z} \exists g_n \forall p, \Pi : \sec_{g_n}(\Pi) > 0$$

$$\exists \varepsilon_0 > 0 \forall p, \Pi \neq \forall p, \Pi \exists \varepsilon_{p, \Pi} > 0$$

- Finite label enumeration is conditional bookkeeping, not geometric classification.
- Uniformity is required over points and planes, not over unnecessary auxiliary choices.

Source and evidence

checks/topology.py
 checks/topology_audit.md
<https://arxiv.org/html/math/0610349v1#S4>

CHAPTER 04

The proof dependency chain

The main argument has several distinct links. First construct a smooth carrier with two commuting group actions and identify its star quotient using marked disk coordinates. Next build southern and northern metrics whose star-horizontal planes have positive curvature. Then compare their boundary metrics in one common gauge and prove a positive sum of outward second fundamental forms. Finally smooth the joined metric while retaining positive sectional curvature and identify the resulting closed smooth manifold.

Each link has a different verification method. Quaternion identities and local tensor formulas admit exact symbolic checks. Norm estimates reduce to algebra plus established inequalities. Smoothness, freeness, and the topology of the quotient require global reasoning. Gluing uses both exact interpolation identities and analytic statements about weak derivatives and mollification. A valid dependency chain cannot replace a missing link by a successful computation elsewhere.

The package consequently records both forward implications and their premises. For example, the northern calculation proves a positive estimate conditional on its profile, cutoff, and boundary data; it does not prove the southern theorem. The gluing calculation assumes the two positive disk metrics and the correct boundary isometry already exist. The final classification invokes the attaching-map identification, not just a boundary metric formula. Reading the chain in this order prevents local evidence from being promoted into an unsupported global conclusion.

carrier $\longrightarrow$ fillings $\longrightarrow$ boundary $\longrightarrow$ gluing

local identity + verified premises $\implies$ usable lemma

- Check every premise at the point where the corresponding result is used.
- A later PASS cannot repair a missing earlier geometric hypothesis.

Source and evidence

docs/CLAIM_MAP.md
 checks/topology_audit.md
 checks/gluing_audit.md

CHAPTER 05

Quaternion algebra and smooth topology

The carrier uses quaternion multiplication, conjugation, and a smooth Blakers-Massey map b . Exact algebra checks verify norm and conjugation identities, quotient-coordinate invariance, and the identities supporting the attaching map. Clearing denominators is legitimate only on the stated nonzero or unit-quaternion domain. The program records that domain rather than interpreting a cleared polynomial identity at points where the original expression was undefined.

The displayed scalar formula for b appears singular at endpoint values of the radial variable. Sage checks the endpoint limits, but a finite limit is not a smoothness proof. The manuscript's power-series argument near one endpoint and smooth division by a simple zero near the other establish all-orders smooth extension. Equivariance then extends through the removable sets. This analytic reasoning is indispensable even when the symbolic limits agree.

For the star quotient, every orbit has a unique representative with fibre coordinate one, giving an explicit disk coordinate and inverse. The marked boundary transition is not the identity in the two extending disk charts. Equivariance makes b constant along the relevant conjugation orbit, and an ordinary induction then gives the formula for every positive and negative integer iterate. The external theorem identifies the smooth type represented by that iterate. None of these global assertions follows from finite residue counting alone.

$$\sigma(x) = b(x)^{-1}xb(x), \quad \sigma^n(x) = b(x)^{-n}xb(x)^n$$

$$y = u^{-1}xu$$

- Two symbolic limits support, but do not replace, smooth-extension arguments.
- The principal-bundle label and the star-quotient smooth type are different data.

Source and evidence

checks/topology.py
 checks/topology_audit.md
 paper/submitted.tex: Section 2 and Appendix A

CHAPTER 06

Curvature conventions and sign tests

Sign conventions are structural inputs to a curvature calculation. The manuscript uses the sectional numerator obtained by applying $R(A,B)$ to B and taking its inner product with A . The source code reconstructs curvature with that convention. A formula computed under the opposite convention could be internally consistent yet reverse every positivity conclusion, so the convention must be checked before inequalities are discussed.

The principal right action produces fundamental fields q mapped to $q e_a$. Direct coordinate differentiation verifies their bracket sign. Differentiating adjoint equivariance produces a second sign in the vertical derivative of the connection curvature. These two operations are separately checked by quaternion component calculations; the expected signs are not merely inserted into a final asserted tensor. A deliberate wrong-sign mutation in one mixed curvature block is also rejected.

Boundary normals have an independent sign issue. The northern outward normal points in the increasing northern radial direction, whereas the southern outward normal is the negative of the southern radial coordinate direction. The fibre slopes therefore cancel in the sum of outward forms, while the angular terms add positively. Confusing the two normals would change the gluing hypothesis. The audit follows the same convention from the source metric derivatives through the quotient restriction and into the external gluing theorem.

$$\sec(A \wedge B) = \frac{\langle R(A, B)B, A \rangle}{|A \wedge B|^2}$$

$$\nu_N = \partial_s, \quad \nu_S = -\partial_t$$

- Verify conventions before comparing tensor components from different sources.
- Negative controls test a specific error-detection path, not all possible sign errors.

Source and evidence

checks/connection.py
checks/connection_audit.md
checks/gluing_audit.md

CHAPTER 07

The complete connection-metric tensor

The southern metric requires more than pure horizontal and pure vertical sectional curvatures. Arbitrary two-planes mix these directions, so the verification reconstructs every relevant curvature block. Starting from the principal frame brackets and curvature-equivariance rule, the program applies Koszul's formula and then differentiates the connection coefficients. The coefficient domain is exact: a rational Laurent polynomial ring with symbolic curvature, derivative, radius-gradient, and Hessian entries.

The calculation uses the actual seven-dimensional base and three-dimensional fibre. It performs 5,176 exact component comparisons: 594 Koszul coefficients, 441 HVHV, 441 HHVV, 189 HVVV, 1,029 HHHV, 2,401 horizontal connection corrections, and 81 vertical components. All checked residuals vanish. The separate quaternion sign calculation adds 45 comparisons. These are executable tensor checks, not 5,176 independently packaged certificates or 5,176 new mathematical theorems.

Using normal frames does not assume that the base is flat. The background base-curvature tensor is restored in the purely horizontal block; the computed correction records the connection contribution. Likewise, the geometric bracket and equivariance identities are inputs whose validity must be read in the manuscript. The value of reconstruction is that the displayed tensor is derived from lower-level local data, making an accidental transcription of the final formula less likely.

$$2\Gamma_{AB}^C = C_{AB}^C - C_{BC}^A + C_{CA}^B$$

$$R_{ABCD} = e_A \Gamma_{BC}^D - e_B \Gamma_{AC}^D + \Gamma_{BC}^E \Gamma_{AE}^D - \Gamma_{AC}^E \Gamma_{BE}^D - C_{AB}^E \Gamma_{EC}^D$$

- All mixed blocks matter when the target quantifies over arbitrary planes.
- Normal-frame reconstruction still relies on geometric input identities.

Source and evidence

checks/connection.py
 checks/connection_audit.md
 certificates/connection.json

CHAPTER 08

Decomposable planes and the master contraction

Sectional curvature is tested on decomposable bivectors: those obtained from two actual tangent vectors. It is not the same condition as positivity of the curvature operator on every element of the exterior square. The proof respects this distinction by expanding an arbitrary pair $X+U$ and $Y+V$, with horizontal components X,Y and vertical components U,V .

Their horizontal, mixed, and vertical bivector components satisfy Plücker relations. The code checks 441 indexed instances of the relation linking these components, together with the full bivector norm decomposition. These identities justify combining curvature blocks on actual planes. They do not assume a curvature-operator lower bound stronger than the base sectional-curvature hypothesis.

A particularly sensitive term is linear in the connection curvature and in the vertical Lie bracket. Direct symbolic contraction gives a contribution of minus one half of J from the mixed diagonal block and minus J from the horizontal-vertical cross block. Their sum is minus three halves of J . The program also verifies the quaternion bracket norm identity, which converts this term into a product of area norms. This is where an omitted cross block or a mistaken factor of two could invalidate an otherwise plausible positivity estimate.

$$\alpha = X \wedge Y, \quad \beta = X \otimes V - Y \otimes U, \quad \gamma = U \wedge V$$

$$\beta_{ia}\beta_{jb} - \beta_{ib}\beta_{ja} = \alpha_{ij}\gamma_{ab}$$

$$J = \langle \Omega(X, Y), [U, V] \rangle$$

- The estimates are for the decomposable cone, not an unjustified positive curvature operator.
- Both the diagonal and cross-block contributions are included in the bracket term.

Source and evidence

checks/connection.py
 checks/connection_audit.md
 paper/submitted.tex: master curvature estimate

CHAPTER 09

Young inequalities and remaining positive reserves

After the tensor contraction, several mixed products must be absorbed into positive squared area terms. This step can fail even if every displayed curvature block is correct: using the same positive reserve twice, missing a coefficient, or ignoring a parameter hypothesis can destroy the final lower bound. The package therefore checks the absorption arithmetic separately.

Young's inequality is represented by an exact completed-square identity after denominators are cleared under the stated positivity assumptions. The code tracks the particular reserve assigned to each loss. It also compares sharper intermediate bounds with the manuscript's deliberately larger common constant $C=4$. The difference has nonnegative coefficients on the stated nonnegative orthant, so the larger printed constants are conservative rather than erroneous.

The final horizontal, mixed, and vertical reserves are the displayed positive coefficients. Their truth is conditional on finite geometric constants, uniform concavity, and the specified smallness inequalities. Polynomial nonnegativity of a slack cannot establish those geometric hypotheses. The parameter module supplies a finite collection of sufficient upper bounds on epsilon after the constants have been fixed; the manuscript explains why those constants exist. The reader should follow both parts rather than reading the reserve identity as an unconditional positivity certificate.

$$px^2 - axy + \frac{a^2y^2}{4p} = \frac{(2px - ay)^2}{4p}, \quad p > 0$$

$$\mathcal{K} \geq \frac{\kappa}{2}h^2 + \frac{\varepsilon\Lambda}{8}m^2 + \frac{1}{8\varepsilon}k^2$$

- Every use of a positive reserve is charged once in the coefficient ledger.
- Exact nonnegative-coefficient checks require the declared orthant assumptions.

Source and evidence

checks/connection.py
 checks/parameter_logic.py
 certificates/connection.json

CHAPTER 10

The southern connection and its radius

The southern filling is built on a round base region with a connection chosen through a cutoff and the transition map g_n . Its potentials are written in two gauges. The program verifies their gauge transformation using noncommutative quaternion multiplication, rather than treating the factors as commuting scalars. Maurer-Cartan reduces the curvature expression to a cutoff-derivative term and a quadratic connection term.

The radius is an actual smooth function of epsilon and the radial coordinate, not merely a formal Taylor polynomial. Symbolic differentiation verifies its logarithmic derivative and second derivative. The round Hessian of the chosen function ϕ has the stated radial and tangential eigenvalues. On the southern region the negative Hessian has the required uniform lower bound after choosing A_0 . The outward boundary normal then determines the negative signs in the southern second fundamental form.

At the south pole, the extending gauge has zero connection potential and the radius is even in the polar variable. These facts support smooth extension, but a symbolic derivative alone is not the complete polar-smoothness theorem. Similarly, finiteness of the connection bounds comes from smooth geometry on a compact region. The Sage calculation verifies the formulas and coefficient implications after those premises are supplied; it does not replace the global construction of the connection or its equivariance.

$$A_N = \chi g_n^{-1} dg_n, \quad A_S = (\chi - 1) dg_n g_n^{-1}$$

$$r = \sqrt{\varepsilon} e^{-\varepsilon A_0 \cos t/2}, \quad \frac{r'}{r} = \frac{\varepsilon A_0 \sin t}{2}$$

- The gauge calculation preserves quaternion multiplication order.
- Smooth extension and finite uniform connection bounds remain geometric arguments.

Source and evidence

checks/connection_audit.md
 checks/connection.py
 paper/submitted.tex: Section 4

CHAPTER 11

The northern star-horizontal graph

The northern metric is a product in the principal-bundle trivialization, but the planes relevant for the quotient are star-horizontal, not principal-horizontal. Their fibre component is determined by the infinitesimal conjugation map K_x . The exact polynomial Gram identity proves its operator norm is at most two throughout the sphere, including points where the conjugation orbit changes dimension.

In orthonormal angular and fibre coordinates, the star generators are $(F K_x \xi_i, r \xi_i)$. Orthogonality gives a graph over the radial and angular base directions. Because the fibre radius is positive, the full orbit Gram matrix remains positive definite even when K_x loses rank. The formula uses the adjoint of K_x , not an inverse; no constant-rank assumption is hidden in the argument.

Two graph estimates control radial-fibre and mixed angular-fibre areas by projected base areas. The mixed estimate follows from an area-preserving orthogonalization and an exact tensor norm expansion. If the angular vectors are dependent, both mixed sides vanish. If both projected base-area squares vanish for a horizontal pair, graph injectivity makes the original pair dependent. Thus positive coefficients for these two areas control every actual horizontal two-plane, including oblique planes and planes whose position varies with the parameters.

$$U = TX, \quad T = -\frac{F}{r} K_x^*, \quad \|T\| \leq \frac{2F}{r}$$

$$|X \otimes TY - Y \otimes TX|^2 \leq 2\|T\|^2 |X \wedge Y|^2$$

- Principal-horizontal and star-horizontal are different distributions.
- Rank changes of K_x do not create singular star orbits.

Source and evidence

checks/north.py: N01-N03
 checks/north_audit.md
 paper/submitted.tex: horizontal graph

CHAPTER 12

Full ten-dimensional warped curvature

The northern source has dimension ten: one radial direction, six angular directions, and three fibre directions. The new verification reconstructs its entire local curvature operator from the metric two-jet in stereographic coordinates. It does not merely check a representative radial plane or rely on the previous package's lower-dimensional slice calculation.

There are 45 coordinate bivectors. The calculation verifies 45 diagonal entries and shows that all 1,980 off-diagonal entries vanish exactly. The diagonal blocks have multiplicities six, three, fifteen, three, and eighteen, corresponding to radial-angular, radial-fibre, angular-angular, fibre-fibre, and angular-fibre planes. The coefficients are rational functions in six independent symbols for the two warping functions and their first two derivatives.

This diagonal structure is important because it supplies the complete numerator for any pair of vectors. A positive radial coefficient and a positive tangential coefficient alone would not suffice if an uncontrolled mixed block were present. Here the mixed loss is explicitly the angular-fibre coefficient and is subsequently bounded using the horizontal graph. The finite symbolic calculation still assumes the standard metric-jet interpretation and curvature convention; it is evidence for the tensor formula, not a formalization of all differential geometry used to interpret it.

$$G_N = ds^2 + F^2 h_{S^6} + r^2 h_{S^3}$$

$$-\frac{F''}{F}, \quad -\frac{r''}{r}, \quad \frac{1-F'^2}{F^2}, \quad \frac{1-r'^2}{r^2}, \quad -\frac{F'r'}{Fr}$$

- All 45 by 45 curvature-operator entries are checked in the actual source dimension.
- The mixed loss is retained, not inferred harmless from the pure blocks.

Source and evidence

checks/north.py: N04
 checks/north_audit.md
 certificates/north.json

CHAPTER 13

Profiles, center regularity and uniform estimates

The angular profile is chosen before the fibre scale. It is the inverse of a strictly increasing integral, so its length is finite once F_a and δ are fixed. This length can be very large; the proof only requires that it be finite before ϵ is chosen. Differentiation gives the two positive base curvature coefficients, and the inverse's oddness supplies the polar smoothness condition at the center.

The fibre radius is constant near the center and increases with a prescribed small boundary slope. Its second derivative is supported where the angular profile has substantial radial curvature. The graph estimates bound the resulting radial-fibre loss there. Away from that support the loss vanishes. The angular loss is controlled on the entire interval by the elementary hyperbolic-sine inequality, leaving at least half the positive angular coefficient.

The final quotient lower bound also accounts for distortion between base area and horizontal area. The graph's exterior-square norm can enlarge a bivector, so the source coefficient bound must be divided by the appropriate graph factor. This is a uniform estimate on all normalized horizontal planes, not a claim obtained by following each fixed plane separately. At the center the smooth limits are positive and the star-horizontal space is the whole base tangent space.

$$F' = e^{-F^2/(2\delta^2)}, \quad F(s) = s - \frac{s^3}{6\delta^2} + O(s^5)$$

$$c_N = \left(1 + \frac{16F_a^2}{r_a^2}\right)^{-2} \min \left\{ \frac{e^{-F_a^2/\delta^2}}{\delta^2}, \frac{1 - e^{-F_a^2/\delta^2}}{2F_a^2} \right\}$$

- Fix the finite angular geometry first; choose ϵ afterwards.
- Area distortion is included in the lower bound for normalized quotient planes.

Source and evidence

checks/north.py: N05-N12
 checks/interval_bounds.py
 paper/submitted.tex: northern lower bound

CHAPTER 14

Boundary isometry and the positive shape sum

Both source boundary metrics have the same product form in the common north gauge. Quotienting by the same star action therefore gives the same boundary metric there. This common gauge does not extend over both disk centers, so equality in that gauge does not make the marked attaching map topologically trivial. In the two extending disk coordinates, the same identification is represented by sigma to the nth power.

The quotient differential in orthonormal components produces the cometric directly as L^* . The generic matrix check uses an arbitrary six-by-three K , verifying the factors and signs independently of a special orbit. The second fundamental forms are computed on the same horizontal lift of a boundary vector. The opposite outward-normal directions make the fibre terms cancel exactly.

The remaining shape sum is strictly positive because μ_N is positive, μ_S is negative, and a nonzero graph vector has nonzero angular component. Its eigenvalues relative to the actual boundary metric include a graph normalization factor. The code checks the gap between these eigenvalues and the announced lower bound and verifies the point where the bound is attained. This is a genuine geometric gluing margin, conditional on the two fillings and the global marked identification having been constructed as stated.

$$h^{-1} = F_a^{-2} I + r_a^{-2} K_x K_x^*$$

$$B_N + \phi_n^* B_S \geq \frac{r_a^2(\mu_N - \mu_S)}{r_a^2 + 4F_a^2} h$$

- Common-gauge metric equality does not erase the nontrivial attaching map.
- The fibre cancellation and the angular positive jump use the same outward-normal convention.

Source and evidence

checks/north.py: N02 and N11
 checks/topology_audit.md
 paper/submitted.tex: Section 6

CHAPTER 15

Gluing interpolation in the actual dimension

The strict gluing argument begins with two already positive metrics and a boundary isometry whose outward second fundamental forms have positive sum. These are substantive hypotheses, not outputs of the interpolation formula. With the normal coordinate increasing across the interface, the derivative jump equals twice the shape sum. Its sign makes the interpolated radial curvature large and positive.

The code verifies the cubic Hermite endpoint jets and their rescaled derivative expansions. It then reconstructs the collar curvature using independent metric jets in the actual ambient dimension seven. The 3,966 exact residuals comprise 2,401 coordinate-curvature comparisons, 36 radial-block comparisons, 1,296 tangential Gauss comparisons, 216 mixed Codazzi comparisons, and seventeen interpolation or algebra identities. Nine unexpanded arithmetic certificates are separately replayable.

The tangential curvature retains a reserve, while the mixed Codazzi block remains bounded and contains no second normal derivative. An exact completed-square estimate controls every combination of a tangential and a radial direction. Thus the permissible collar width can be chosen uniformly over all plane angles, including angles that depend on the width. This avoids the common error of proving positivity only for pure radial and pure tangential planes while leaving oblique planes uncontrolled.

$$\Delta = P_0 - Q_0 = 2(B_N + \phi_n^* B_S) > 0$$

$$R_{izzj} = -\frac{1}{2}H''_{ij} + \frac{1}{4}H'_{ik}H^{kl}H'_{lj}$$

- The seven-dimensional reconstruction checks mixed as well as pure curvature blocks.
- The gluing lemma does not assert arbitrary positive-curvature connected-sum preservation.

Source and evidence

checks/gluing.py
 checks/gluing_audit.md
 paper/submitted.tex: Appendix B

CHAPTER 16

Why smoothing still needs human analysis

The interpolated collar is initially piecewise smooth with matching first jets. That regularity makes it locally $C^{1,1}$ and ensures that its second weak derivatives have no Dirac mass at the joining interfaces. Sage checks the endpoint identities and zero jump coefficients. Distributional differentiation and the regularity conclusion themselves are analytic arguments, not consequences certified by those polynomial identities alone.

The smoothing uses a single nonnegative convolution kernel in the normal direction, acting on tensor values in one fixed tangent-space bundle over the boundary. This is not arbitrary componentwise averaging across changing charts. The all-lowered curvature tensor splits into a constant-coefficient linear second-derivative term and a nonlinear expression continuous in the metric and its first derivatives. Convolution commutes with the former; uniform C^1 convergence controls the latter.

The proof averages sectional numerators on the same decomposable bivector and compares the nearby metric norms uniformly. It does not assume positivity of the entire curvature operator. The collar width is fixed first, producing a positive margin; the smoothing width is chosen afterwards. Cutoff localization occurs away from the nonsmooth interface, where ordinary C^2 convergence is available. These convergence and compactness arguments are marked UNVERIFIED by computation even though the manuscript supplies mathematical proofs and the scoped audit found no gap.

$$R(G) = \mathcal{L}(D^2G) + \mathcal{Q}(G, DG)$$

τ fixed first, $\nu \rightarrow 0$ afterwards

- C^1 convergence controls the nonlinear error; false C^2 convergence at the seam is not assumed.
- UNVERIFIED here records analytic scope, not a discovered failure.

Source and evidence

checks/gluing_audit.md

checks/gluing.py: analytic dependency groups

paper/submitted.tex: tensor-valued smoothing

CHAPTER 17

A noncircular order for all parameters

Small-parameter arguments require a dependency order. For a fixed integer n , choose the cut, cutoff, and southern connection first. Their finite curvature and derivative bounds determine a permissible concavity scale and A_0 . With the same boundary data, choose the northern cutoff and delta, then the angular profile and its finite length. Only after these quantities are fixed should the common epsilon be selected.

The parameter module records a finite list of explicit positive upper bounds sufficient for the algebraic smallness requirements. Their minimum is positive because every denominator involves fixed finite constants and the required positive geometric inputs. This is an existence argument for a common range, not optimization of the largest allowable epsilon. The northern conditions introduce no lower bound that conflicts with the southern upper bounds.

Once epsilon is selected, both disk metrics and their strict curvature and boundary margins are fixed. The gluing width is then chosen small enough for the mixed-plane estimates, and the smoothing width is chosen last. Constants may deteriorate sharply with earlier choices, especially with the northern interval length. That deterioration is harmless for existence if it is finite and the order is respected. It would be invalid to shrink one parameter while silently treating quantities that depend on it as fixed.

$$n \longrightarrow A_0, \Lambda \longrightarrow \delta, \ell_N \longrightarrow \varepsilon \longrightarrow \tau \longrightarrow \nu$$

$$0 < \varepsilon < \min \{ \varepsilon_S, \varepsilon_N \}$$

- A positive minimum of finitely many fixed upper bounds is enough for existence.
- The construction does not require a numerically convenient or optimal parameter.

Source and evidence

checks/parameter_logic.py
 checks/north_audit.md
 checks/gluing_audit.md

CHAPTER 18

What the rigorous interval calculation covers

The new interval module uses Sage RealIntervalField at 200-bit precision with outward rounding. It does not evaluate ordinary floating-point samples and infer positivity between them. Instead, it partitions the complete interval from one half to four into 224 closed rational cells of width one sixty-fourth. For each cell, interval arithmetic encloses every value of the hyperbolic-sine ratio and proves that its lower endpoint exceeds one half.

Every input-cell endpoint and every output-enclosure endpoint is exported as an exact rational string. The unbounded domain is not falsely described as covered by finitely many cells. The positive-term series gives an analytic lower bound of two on the small tail and eight thirds on the large tail. These tail arguments are recorded separately from the interval computation. Additional enclosures check strict scalar reserves and one concrete northern-only parameter witness.

For that witness, the program encloses the finite length bound, the fibre slope, and positive scalar c_N and c_B . The chosen A_0 is not claimed to satisfy the southern connection requirements for any n . Therefore this witness is not an explicit closed-manifold metric and does not certify all 28 types. Reproducing the transcendental interval calculations requires Sage and its interval engine; the standard-library rational checker cannot recompute exponential or hyperbolic-sine enclosures.

$$R(x) = \frac{2\sinh(x^2/2)}{x^3} > \frac{1}{2}$$

$$0 < x \leq \frac{1}{2} : R(x) \geq 2, \quad x \geq 4 : R(x) \geq \frac{8}{3}$$

- The 224 cells cover a scalar interval, not the manifold or its plane bundle.
- Exact endpoint recording is distinct from independent transcendental recomputation.

Source and evidence

checks/interval_bounds.py
 certificates/interval_bounds.json
 checks/north_audit.md

CHAPTER 19

The exact-certificate formats

An exact certificate is a finite object that exposes a mathematical equality or inequality in a form another program can inspect. The package distinguishes sparse polynomial identities, unexpanded arithmetic-expression identities, and polynomials with nonnegative coefficients on a declared orthant. The distinction matters because these formats provide different amounts of independent reconstruction.

A sparse identity compares rational coefficient dictionaries attached to monomials. An expression certificate instead supplies the two arithmetic expressions; the independent checker expands them using rational arithmetic and compares the results. The latter can catch errors in expansion as well as disagreement between supplied coefficient lists. A nonnegative-coefficient certificate proves nonnegativity only when every declared variable has the required nonnegative value; it is not a universal positivity claim over arbitrary real inputs.

Certificate files also contain contextual information and Sage-only evidence that the rational checker does not independently verify. Thousands of curvature-jet residuals are reproduced by running the Sage modules. Interval enclosures require their dedicated interval module. Human-readable explanations specify how the checked algebra enters a geometric proof. A public reader should inspect each certificate's identifier, source correspondence, variables, assumptions, and replay mechanism rather than treating the whole JSON file as a single opaque seal of correctness.

$$P = \sum_{\alpha} c_{\alpha} X^{\alpha}, \quad c_{\alpha} \in \mathbb{Q}$$

$$c_{\alpha} \geq 0, \quad x_i \geq 0 \implies P(x) \geq 0$$

- Unexpanded expressions give stronger independent reconstruction than two pre-expanded lists.
- Domain assumptions accompany every polynomial positivity conclusion.

Source and evidence

certificate_format.py
 verify_certificates.py
 docs/MATHEMATICAL_CERTIFICATES.md

CHAPTER 20

Independent replay and deliberate corruption

The independent certificate checker uses only the Python standard library, with `Fraction` for exact rational arithmetic. It has no Sage dependency and does not invoke a symbolic algebra service. Its restricted arithmetic parser accepts only the intended names and operations, rejects unsupported syntax, and expands the permitted expressions itself. This reduces dependence on the same algebra engine that produced the certificates.

The recorded run successfully replays 48 certificates. A separate negative-control stage deliberately changes the right-hand side of every eligible expression certificate by adding one. All 33 such altered certificates are rejected. This demonstrates that the tested identity path is sensitive to a real mathematical discrepancy rather than accepting every submitted object. The connection module additionally tests a specific wrong bracket sign against its reconstructed tensor.

These controls have limits. They do not prove that the parser, rational arithmetic implementation, or hardware is infallible. They do not show that every possible translation error would be detected. A perfectly replayed identity can still concern the wrong geometric quantity if the source-to-code correspondence is mistaken. The package therefore combines independent replay with source binding, readable formulas, scoped mathematical audits, and raw reproduction logs. Independence is a carefully described layer of evidence, not a synonym for complete formal verification.

$$P = Q \quad \longrightarrow \quad P = Q + 1 \text{ must fail}$$

rational replay $\neq$ geometric interpretation

- Mutation rejection tests the actual checker, not just the producer.
- Inspect source correspondence even when all exact identities replay.

Source and evidence

verify_certificates.py
 verify_all.py
 reproduction/reference-run/logs

CHAPTER 21

Reproduction and the recorded environment

The reference run uses the locally installed SageMath 10.9 environment. The package records software versions, platform information, execution times, source and code hashes, and raw module outputs. Environment files describe both a portable dependency request and the concrete package set used for the reference machine. They are evidence about the executed setup, not a guarantee that every future solver or platform will behave identically.

The normal reproduction entry point runs the modules, collects their structured output, invokes certificate replay, and executes the negative controls. A failed process, malformed result, missing output, or failed check must not be converted into a passing result. A fresh output directory separates a reader's run from the archived reference run. The immutable source and accompanying checksum manifest make accidental replacement detectable.

A useful reproduction proceeds in two stages. First run the independent rational checker in a plain Python environment; this tests the portable certificate layer without Sage. Then run the complete Sage pipeline to reconstruct curvature tensors and interval enclosures. Compare the mathematical statuses, exact residuals, source hashes, and certificate contents. Wall-clock time can differ across machines and is not a correctness criterion. The package does not embed an entire multi-gigabyte Sage installation; it provides the information needed to install or select a compatible one.

source identity + environment + command + logs

reference run $\neq$ reader run

- Preserve the archived reference output; write a new local-run directory.
- Compare exact mathematical results rather than runtime or superficial log order.

Source and evidence

```
reproduce.sh
environment.yml
reproduction/conda-packages.json
reproduction/reference-run/environment.json
```

CHAPTER 22

External classification and geometric theorems

Some essential inputs are external mathematical results. The Durán-Püttmann-Rigas theorem and its attaching-map description identify the oriented smooth type associated with the n th iterate of the specified diffeomorphism. The source comparison checks that the manuscript uses the same map and orientation convention. It is not enough to cite a theorem about a similarly named construction.

Reiser-Wraith supplies a gluing theorem for isometric compact boundaries and a nonnegative sum of second fundamental forms. The sectional-curvature case corresponds to its first intermediate Ricci curvature. Its quantitative corollary permits a strict curvature threshold, which is the cited justification for the paper's final bound by one half of the smaller disk bound. The manuscript has a stronger positive shape sum, and its normal convention must agree with that external theorem.

O'Neill's submersion formula is another structural input. For orthonormal horizontal lifts, it transfers a lower bound on the relevant source planes to the quotient because the additional term is nonnegative. The principal and star submersions must not be confused when applying it. The audits checked these interfaces and found the cited statements applicable within their recorded scope. Reading and checking applicability is different from re-proving the external theorems, and neither the Sage run nor the rational certificates formalize them.

$$\Sigma_n \leftrightarrow n \bmod 28$$

$$\text{sec}_{\text{quotient}} = \text{sec}_{\text{horizontal source}} + \frac{3}{4} |[X, Y]^{\text{vert}}|^2$$

- External results must match the actual map, convention, and hypotheses.
- The precise final quantitative bound uses the cited gluing corollary.

Source and evidence

<https://arxiv.org/html/math/0610349v1#S4>
<https://arxiv.org/html/2308.06996v2>
 checks/topology_audit.md
 checks/gluing_audit.md

CHAPTER 23

What PASS does not establish

The package is not a Lean development and does not contain a formal proof of the complete main theorem. It does not classify smooth homotopy spheres from first principles, construct every smooth chart inside a proof assistant, or formalize the submersion and smoothing theorems. It also does not numerically subdivide every point and every plane of an explicit closed metric.

A passing group certifies the operation described in that group's evidence. For example, exact agreement of the reconstructed northern curvature operator is strong evidence for the printed local formula. It is not, by itself, evidence that the quotient has the claimed global smooth type. Likewise, successful gluing identities cannot establish that the two disks satisfy their boundary hypotheses. These limitations are logical distinctions, not disclaimers that can be discarded once a run is green.

The scoped audits found no substantive defect in the portions they reviewed. That statement should not be promoted into an independent external referee decision or a guarantee that no error exists. The public claim supported by the package is narrower and more useful: specified computations were actually run, their exact outputs are reproducible, deliberately corrupted certificates are rejected, and the remaining analytic and external dependencies are visibly separated. A mathematical reader can then focus attention on the parts that calculation does not settle.

PASS(C) $\implies$ the stated check C succeeded
not machine proved $\neq$ disproved

- Do not advertise the all-types theorem as fully machine-certified.
- An honest dependency list improves the package's scientific usefulness.

Source and evidence

docs/PROOF_SCOPE.md
 VERIFICATION_REPORT.md
 reproduction/reference-run/summary.json

CHAPTER 24

A reviewer runbook and responsible publication

Start by confirming the manuscript version and checksum, then read the main theorem and the dependency map. Select a load-bearing identity, locate its source formula and program derivation, and replay its certificate independently. Run the complete Sage pipeline next, checking that the raw module outputs agree with the recorded statuses. This sequence tests identity, interpretation, and execution separately.

The mathematical review should then concentrate on the non-computational links: smooth extension at the carrier endpoints, marked disk and seam identifications, equivariance of the constructed connection, common parameter order, global all-plane estimates, and analytic smoothing. Check the external classification and gluing statements at their primary sources. If a discrepancy is found, record whether it is a source error, a translation error, an implementation error, an unsupported dependency, or only an expository issue; these require different repairs.

A public verification website should make downloads, source hashes, reproduction commands, result counts, and scope limits easy to find. Attractive presentation must not replace the distinction between exact replay, interval recomputation, and human analysis. Releases should be versioned so that a reader can cite the precise package used. Correcting the paper or code should produce a new recorded run rather than overwriting historical evidence. The strongest public contribution is a clear, inspectable path from a stated claim to the calculations and remaining arguments that support it.

identify → inspect → replay → review

new source or code ⇒ new recorded run

- Publish reproducible evidence together with its exact verification boundary.
- Report discovered discrepancies by type and preserve the earlier release.

Source and evidence

README.md
SHA256SUMS
checks/*_audit.md
reproduction/reference-run

APPENDIX A / 1-8

Complete check register

Claims are quoted from the executed checks. Evidence below is abbreviated; summary.json retains complete objects. PASS does not establish upstream geometric assumptions.

S01 / PASS / source_binding

The submitted source is exactly the reviewed v12 file, not a previous manuscript

```
{"actual_sha256": "ea27de068cd78871211d0a1ac102e76b65c7bd1f033267fb24a2c5f8dc423f9e",
"reviewed_sha256": "ea27de068cd78871211d0a1ac102e76b65c7bd1f033267fb24a2c5f8dc423f9e"}
```

S02 / PASS / source_binding

Every inventoried labeled display matches the separately frozen source contract

```
{"display_blocks": 80, "labels": 92}
```

S03 / PASS / source_binding

This v12 release does not silently change the supplied manuscript

Both source copies are preserved byte-for-byte.

T01 / PASS / topology

Quaternion multiplication is associative

Four polynomial identities over $\mathbb{Q}\mathbb{Q}$ in 12 independent real variables.

T02 / PASS / topology

Quaternion norm is multiplicative and conjugation reverses products

Hamilton multiplication was expanded from components; no sampled values.

T03 / PASS / topology

The quotient coordinate $u^{-1} \times u$ is invariant under $(x,u) \rightarrow (q \times q^{-1}, qu)$

Four generic denominator-cleared polynomial identities; nonzero norms required for inverses.

T04 / PASS / topology

Bundle transition and right principal action commute

Associativity applied to $g_n(x)$, u , h ; smoothness of g_n is a separate analytic premise.

T09 / PASS / topology

The marked transition is conjugation by $g_n(y)$, after imposing the equivariance premise

Four exact denominator-cleared identities in arbitrary quaternion components; u and $g_n(x)$ are unit in the geometric application.

APPENDIX A / 9-16

Complete check register

T10 / PASS / topology

Opposite conjugations are inverse when the conjugating quaternion is unit

The polynomial identity is $C_b(C_{\bar{b}}(x))=|b|^4 x$. The manifold inverse formula additionally uses $b(\sigma(x))=b(x)$, proved using equivariance in the manuscript.

T11 / PASS / topology

The induction step for iterated conjugation has the required multiplication order

A generic two-step conjugation identity, not sampling of the nonlinear Blakers--Massey map. The all-integer induction is human reasoning from this group law and invariance of b along σ -orbits.

T05 / PASS / topology

$$K^* K = 4(|p|^2 + |\operatorname{Im} w|^2)I - p^T \operatorname{Im}(w) \operatorname{Im}(w)^T$$

Exact 3-by-3 matrix identity.

T06 / PASS / topology

On $|x|=1$, $\|K_x v\|^2 \leq 4\|v\|^2$, including all orbit ranks

The deficit is the displayed sum of squares, valid for every real vector v .

T07 / PASS / topology

Blakers--Massey scalar factor has endpoint limits π and $\pi/2$

Exact symbolic limits. Limits alone are not an all-orders smoothness proof.

T08 / PASS / topology

Residues $0, \dots, 27$ give 28 labels and orientation reversal gives 15 orbits

Finite modular bookkeeping only; the geometric interpretation uses DPR Theorem 1.

connection.v12_source_binding / PASS / connection

Audited connection formulas are bound to the submitted v12 source

```
{ "audited_sha256": "ea27de068cd78871211d0a1ac102e76b65c7bd1f033267fb24a2c5f8dc423f9e", "label_lines":
{ "eq:HHHH": [1031], "eq:HHHV": [989], "eq:HHVV": [979], "eq:HVVH": [974], "eq:HVVV": [983], "eq:Kosz...
```

connection.right_action_sign / PASS / connection

Quaternion vector fields independently confirm bracket and equivariance signs

```
{ "bracket": "[F_a, F_b]=F_[e_a, e_b]", "equivariance_derivative": "d(Ad(exp(-t e_a))W)/dt=-[e_a, W]",
"exact_component_checks": 45, "fundamental_field": "F_a(q)=q e_a" }
```

APPENDIX A / 17-24

Complete check register

connection.koszul / PASS / connection

Koszul reconstruction agrees with Koszul formula

```
{"base_dimension": 7, "exact_component_comparisons": 594, "failures": []}
```

connection.hvhv / PASS / connection

Koszul reconstruction agrees with HVHV formula

```
{"base_dimension": 7, "exact_component_comparisons": 441, "failures": []}
```

connection.hhvv / PASS / connection

Koszul reconstruction agrees with HHVV formula

```
{"base_dimension": 7, "exact_component_comparisons": 441, "failures": []}
```

connection.hvvv / PASS / connection

Koszul reconstruction agrees with HVVV formula

```
{"base_dimension": 7, "exact_component_comparisons": 189, "failures": []}
```

connection.hhhv / PASS / connection

Koszul reconstruction agrees with HHHV formula

```
{"base_dimension": 7, "exact_component_comparisons": 1029, "failures": []}
```

connection.hhhh / PASS / connection

Koszul reconstruction agrees with HHHH formula

```
{"base_dimension": 7, "exact_component_comparisons": 2401, "failures": []}
```

connection.vvvv / PASS / connection

Koszul reconstruction agrees with VVVV formula

```
{"base_dimension": 7, "exact_component_comparisons": 81, "failures": []}
```

connection.curvature_sign_negative_control / PASS / connection

A deliberately reversed HVHV bracket sign is rejected

```
{"deliberate_error": "Replace -<[e_0,e_1],Omega_01>/4 by its positive opposite.", "rejection_residual": "-O0_1_2"}
```

APPENDIX A / 25-32

Complete check register

connection.plucker / PASS / connection

The printed Pluecker identity has the correct sign

```
{"symbolic_identities": 441}
```

connection.bracket_contraction / PASS / connection

Linear curvature terms sum to $-3J/2$

```
{"alpha_gamma": "-J", "beta_beta": "-J/2", "orientation_witness": "Omega_12=e_3, X=e_1,Y=e_2,U=e_1,V=e_2 gives J=2 and total=-3.", "total": "-3*J/2"}
```

connection.horizontal_section / PASS / connection

Derived horizontal correction contracts to $-3r^2|\Omega(X,Y)|^2/4$

```
{"exact_polynomial_residual": "0"}
```

connection.bivector_norm / PASS / connection

$h^2+m^2+k^2=|(X+U) \wedge (Y+V)|^2$

```
{"exact_polynomial_residual": "0"}
```

connection.quaternion_bracket_norm / PASS / connection

The SU(2) bracket bound is an exact norm identity

```
{"fibre_normalization": "[e_a,e_b]=2 epsilon_abc e_c", "identity": "|[U,V]|^2=4|U wedge V|^2"}
```

connection.norm_kernels / PASS / connection

Cauchy--Schwarz and the fibre dimension-three array factor are exact

```
{"HVVV_array": "sum_abc(delta_ab w_c-delta_ac w_b)^2=4|w|^2", "cauchy_sos": "|a|^2|b|^2-<a,b>^2=sum_(i<j)(a_i b_j-a_j b_i)^2", "ordered_skew_pair_norm": "Full squared norm = 2 * restricted squared norm,..."
```

connection.common_constant / PASS / connection

$C=4$ dominates every sharp-master loss

```
{"nonnegative_gap": "13/4*r^2*M0^2*h^2 + 7/2*r^2*M0^2*m^2 + 3*r^2*M0^2*h*k + 2*r*M0*T*m*k + 2*r*M1*h*m + M0*h*k"}
```

connection.young_identity / PASS / connection

Every absorption uses a certified nonnegative square

```
{"identity": "p x^2 + a^2 y^2/(4p)-a x y = (2p x-a y)^2/(4p), p>0"}
```

APPENDIX A / 33-40

Complete check register

connection.reserve_arithmetic / PASS / connection

The written losses leave at least the claimed reserve

```
{"claimed_mixed": "1/8", "horizontal_over_kappa": "1/2", "mixed_over_eps_Lambda": "1/8", "vertical_times_eps": "1/8"}
```

connection.young_coefficients / PASS / connection

All three printed Young coefficients are sufficient

```
{"assumptions": ["r^2 <= 2 eps", "T <= eps D0/2", "eps D0 M0 <= 1", "2 eps M0 <= 1"], "hk_required_coefficient": "8 C^2 M0^2/kappa; printed 8", "hm_required_coefficient": "4 C^2 eps/kappa (M1+1)^2; prin..."}
```

connection.smallness_implications / PASS / connection

Each smallness hypothesis supplies its assigned loss budget

```
{"cleared_denominator_identities": 7, "failures": [], "premises": {"derivative_mixed_loss": "kap Lam-64 C^2 (M1+1)^2 >= 0; denominator 8 kap > 0", "hessian_loss": "Lam-4 eps D0^2 >= 0; denominator 16 > ..."
```

connection.radius_chain_rule / PASS / connection

Differentiating $r = \sqrt{\text{eps}} \exp(\text{eps} \phi/2)$ gives the claimed Hessian tensor

```
{"formula": "-Hess(r)/r = -eps Hess(phi)/2 - eps^2 dphi tensor dphi/4", "verification": "Symbolic arbitrary phi(z0,z1), including mixed derivatives, in normal coordinates"}
```

south.radius_derivatives / PASS / connection

Southern r , logarithmic slope and Hessian have the printed signs

```
{"Hess_phi_angular": "A0*cos(t)", "Hess_phi_radial": "A0*cos(t)", "N_radial": "-1/4*A0^2*ee^2*sin(t)^2 - 1/2*A0*ee*cos(t)", "concavity": "On a<=t<=pi with a>pi/2: -A0 cos(t)>=A0|cos(a)|.", "r_prime_over..."}
```

south.pole_and_boundary / PASS / connection

Pole parity, mixed pole coefficient, boundary form and slope scaling agree

```
{"limit_eps_to_zero": "A0 sin(a)/2", "outward_form": "-(cos(a)/sin(a)) |X|^2 - (eps A0 sin(a)/2) |U|^2", "polar_radius": "sqrt(ee)*e^(1/2*A0*ee*cos(rho))", "pole_N": "eps A0/2", "q_s_over_r_a_squared": ...}
```

south.connection_curvature / PASS / connection

Maurer--Cartan gives the stated cutoff curvature

```
{"curvature": "chi^2*theta_square - chi*theta_square + dchi_theta", "gauge_transform": "Ad(g^-1)((chi-1) dg g^-1)+g^-1 dg=chi g^-1 dg"}
```

south.noncommutative_gauge / PASS / connection

Quaternion multiplication verifies the southern gauge transformation

```
{"denominator_cleared": "|g|^4, positive for g != 0", "exact_component_checks": 4, "identity": "g^-1 ((chi-1) dg g^-1) g + g^-1 dg = chi g^-1 dg", "limitation": "The actual smooth global g_n and its equ..."}
```

APPENDIX A / 41-48

Complete check register

south.external_geometry / UNVERIFIED / connection

Global marked-disk identification, action freeness, and O'Neill descent are external to this symbolic module
This module certifies the displayed local algebra; it does not replace bundle topology or a proof of the Riemannian-submersion theorem.

N01 / PASS / north

The infinitesimal conjugation map is tangent and $||K_x|| \leq 2$.
Exact polynomial Gram identity and sum-of-squares certificate on $|x|=1$.

N02 / PASS / north

The star-horizontal graph and boundary cometric have the stated factors and signs.
Generic 6 by 3 K; all matrix identities are exact over a rational function field.

N03 / PASS / north

Both area estimates are valid, including the mixed-area factor 8.
Exact tensor, Gram determinant, radial linearity and determinant-one shear identities; apply operator norm after orthogonalization.

N04 / PASS / north

The complete doubly warped curvature operator has exactly the five claimed blocks.
Independently reconstructed from the full 10-dimensional metric 2-jet: 45 diagonal entries match and 1980 off-diagonal entries vanish exactly.

N05 / PASS / north

The profile derivatives, curvature coefficients and center Taylor jet are correct.
Exact symbolic differentiation and vanishing ODE Taylor defect.

N06 / PASS / north

The global elementary lower bound is valid for every $x>0$.
Exact minimizer factorization and rational fourth-power comparison; positive convergent sinh series supplies the preceding inequality.

N07 / PASS / north

The radial criterion and the claimed uniform P bound follow with strict reserve.
Rational positive-coefficient certificate; elementary exponential-series bounds avoid numerical approximations.

APPENDIX A / 49-56

Complete check register

N08 / PASS / north

The angular curvature criterion holds uniformly and leaves at least half the angular coefficient.
Exact coefficient arithmetic combines $\delta^3 A_0 F_a \leq 1/(128(1+C_\eta))$, the density bound and N06.

N09 / PASS / north

The small-parameter northern inequalities and matching boundary data coexist.
An explicit positive ϵ_N is supplied after δ and finite ℓ_N are fixed; no lower bound on ϵ conflicts with the southern upper bounds.

N10 / PASS / north

There is an explicit exact witness for the northern parameter inequalities.
All final comparisons are rational; $e < 4$ gives the finite length and boundary derivative bounds.

N11 / PASS / north

The outward shape terms cancel exactly in the fibre directions; the stated boundary margin is positive and sharp.
Exact rational eigenvalue identity and nonnegative gap factorization for $0 \leq b \leq 4$; sharp eigenvalue 4 verified.

N12 / PASS / north

The announced positive quotient lower bound follows from the verified coefficients.
N07/N08 plus elementary exponential monotonicity and the second exterior-power operator norm inequality.

N13 / UNVERIFIED / north

Global analytic/topological hypotheses needed to turn the exact north certificates into a complete filling and gluing theorem.
Not formalized: smooth η and inverse profile, global quotient marking, earlier southern theorem and the subsequent gluing theorem. These are distinguished from the exact checks above.

gluing.hermite.endpoint_jets / PASS / gluing

Cubic interpolant matches values and first normal derivatives at both interfaces.
Four identities in $QQ(z,t,a,b,p,q)$, with $t \neq 0$; each residual is exactly zero.

gluing.hermite.derivatives / PASS / gluing

Displayed first and second derivatives are exact.
Formal differentiation of the interpolant over the exact rational-function field.

APPENDIX A / 57-64

Complete check register

[gluing.hermite.formal_orders](#) / PASS / gluing

Finite Taylor jets have the claimed scaled leading orders $H=h+O(t)$, $H'=((1-y)P+(1+y)Q)/2+O(t)$, $H''=-(P-Q)/(2t)+O(1)$. Arbitrary independent one-sided normal Taylor jets through degree four; $z=t*y$. All three normalized remainders coerce to a polynomial ring, so have no negative t powers. Uniform smooth-family remainder ...

[gluing.determinant](#) / PASS / gluing

The 2-by-2 determinant interpolation identity has the sign stated in the manuscript.
Universal polynomial identity for two arbitrary symmetric 2-by-2 matrices.

[gluing.tangential_gain](#) / PASS / gluing

Subtracting the interpolated determinant gives the nonnegative tangential correction $+l(1-l) \det(P-Q)/4$ when $0 \leq l \leq 1$ and $P-Q$ is positive definite.
Exact identity; positive definiteness and the interval restriction are explicit mathematical premises.

[gluing.curvature.coordinate_formula](#) / PASS / gluing

The displayed all-lowered coordinate curvature formula has the correct convention and separates linear second derivatives from nonlinear first derivatives.
2401 exact polynomial component identities from independent differentiation of connection coefficients, for arbitrary collar 2-jets in the manuscript's dimension seven at slice normal coordinates.

[gluing.curvature.radial](#) / PASS / gluing

Radial block equals $-H''/2 + H' H^{-1} H'/4$.
36 exact entries; $H=I$ at the evaluation point, so $H^{-1}=I$. The quadratic term is a sum of squares on diagonal entries.

[gluing.curvature.gauss](#) / PASS / gluing

Tangential Gauss block has the sign that yields $K_G = K_H \det(H'|P_i)/(4 \det(H|P_i))$.
1296 arbitrary formal tangential components in dimension seven, independently calculated from the Levi-Civita connection.

[gluing.curvature.codazzi](#) / PASS / gluing

Mixed curvature block equals $((D_j H')_{ik} - (D_i H')_{jk})/2$ and contains no H'' .
216 arbitrary formal mixed components; tangential slice Christoffels vanish at the point, so covariant derivatives there equal partial derivatives.

[gluing.mixed.schur](#) / PASS / gluing

The all-plane lower bound follows by an exact nonnegative square, including angles depending on τ .
Identity for $x=|\cos(\theta)|$, $y=|\sin(\theta)|$ with $a_0 > 0$. $\text{Gap} = (a_0 x - 2 M y)^2 / (2 a_0)$. A sufficient explicit choice is $0 < \tau \leq \beta / (C_0 + 2 M^2 / a_0 + a_0/2)$, also subject to all earlier smallness bounds.

APPENDIX A / 65-72

Complete check register

gluing.interface.distributional_jump_coefficients / PASS / gluing

Jump coefficients of the metric and first derivative vanish at both interfaces.

The four exact zero jumps eliminate delta-prime and delta contributions in a one-dimensional distributional second derivative. Distribution theory itself is an analytic input.

gluing.analytic.uniformity / UNVERIFIED / gluing

Uniform C_X^j estimates, positivity for sufficiently small τ , and compact minima on all plane bundles.

UNVERIFIED by finite algebra. Analytic audit finds the supplied argument valid for fixed smooth families on compact X with $\Delta \geq 2 \cdot b_* \cdot h$ and positive input sectional curvature; these hypotheses are not...

gluing.analytic.smoothing / UNVERIFIED / gluing

Common tensor convolution and localized smoothing preserve positive sectional curvature.

UNVERIFIED by finite algebra. Analytic audit finds no gap: C1 matching removes singular measures;

$R = L(D^2G) + Q(G, DG)$; C1 convergence controls Q ; nonnegative convolution preserves the decomposable-bivector...

gluing.application.input_metrics / UNVERIFIED / gluing

The particular disk metrics satisfy the global positive sectional-curvature and strictly positive boundary-jump hypotheses.

Outside this module: no finite Appendix B identity verifies the earlier geometric construction or its premises.

gluing.external.quantitative_bound / UNVERIFIED / gluing

Section 6 obtains $K > \min(c_S, c_N)/2$ from Reiser-Wraith Corollary B.

Primary source arXiv:2308.06996v2, Theorem A(i) and Corollary B, was read on 2026-09-25: $k=1$ is sectional curvature and the lower threshold may be any real κ . This is a checked external theorem inte...

P01 / PASS / parameter_logic

$A/(B(1+x))$ gives slack $A/(1+x)$ in $A \cdot B \cdot \epsilon \cdot x$ at the threshold

Universal rational identity for $A, B > 0$ and $x \geq 0$; expressions below instantiate it.

P02 / PASS / parameter_logic

For $0 \leq \epsilon \leq 1$, $\epsilon^3 \leq \epsilon$

Exact factorization $\epsilon - \epsilon^3 = \epsilon(1 - \epsilon)(1 + \epsilon)$.

P03 / PASS / parameter_logic

Every displayed scalar upper bound satisfies its corresponding linear constraint

11 exact rational identities. Cubic constraints also use P02. Finiteness and signs of geometric constants are explicit premises.

APPENDIX A / 73-80

Complete check register

P04 / PASS / parameter_logic

Northern $d < 1$ follows with a fixed strict reserve from the chosen thresholds
 $d^2 = \epsilon * (\epsilon \wedge \text{Fa})^2 \exp(\epsilon \wedge \text{c})/4 < \epsilon/8 \leq 1/8$.

P05 / PASS / parameter_logic

The allocated mixed-loss budget leaves exactly $\epsilon \text{Lambda}/8$
 The mixed coefficient budget is exactly $\epsilon \text{Lambda}/8$. Other reserves are $\kappa/2$ and $1/(8 \epsilon)$.

P06 / UNVERIFIED / parameter_logic

The concrete n -dependent curvature suprema and smooth cutoff derivative bounds are finite, and all interval/exponential monotonicity hypotheses apply
 This is established by the smooth compact geometric construction and elementary analysis; no finite CAS assertion formalizes these global statements.

I01 / PASS / interval_bounds

The rational cells cover $[1/2, 4]$ without a gap.
 224 closed rational intervals of width $1/64$, exact adjacency checks.

I02 / PASS / interval_bounds

The elementary ratio is greater than $1/2$ on every point of $[1/2, 4]$.
 All 224 outward-rounded interval lower endpoints are greater than the exact rational $1/2$.

I03 / PASS / interval_bounds

Exact endpoint comparisons close the small and large tails, conditional on the positive sinh-series bound.
 Small tail $R \geq 2$ and large tail $R \geq 8/3$; both exceed $1/2$. Tail reasoning is analytic, not finite sampling.

I04 / PASS / interval_bounds

The stated strict exponential and fourth-root reserves have rigorous interval separation.
 Three outward-rounded enclosures separated from their exact rational thresholds.

I05 / PASS / interval_bounds

The concrete northern-only parameter witness obeys both ϵ conditions.
 Rigorous enclosures show $q_{\text{ell}_N} < 1/2$ and $0 < d < 1$; delta comparisons are exact rational.

APPENDIX A / 81-87

Complete check register

I06 / PASS / interval_bounds

The witness gives strictly positive certified scalar c_N and c_B .

Exact rational lower endpoints of both outward-rounded intervals are positive; no underflow or floating-point sign test.

D01 / PASS / document

Revised LaTeX labels are unique

```
{"duplicates": [], "labels": 115}
```

D02 / PASS / document

All ref/eqref/hyperref targets exist

```
{"missing": []}
```

D03 / PASS / document

All citation keys have bibliography entries

```
{"missing": []}
```

D04 / PASS / document

All nine theorem, lemma and proposition statements agree byte-for-byte with the supplied v12

```
{"entire_source_unchanged": true, "statement_blocks": 9}
```

independent_exact_replay / PASS / certificate_replay

The standard-library-only checker accepts all exported supported certificates

```
{"checked": 48, "failures": [], "scope": "encoded rational polynomial claims only"}
```

incorrect_certificates_rejected / PASS / certificate_replay

Adding 1 to the right side of each unexpanded identity is detected

```
{"accepted_mutations": [], "all_rejected": true, "mutations_tested": 33}
```

APPENDIX B / 1-24

Source formula inventory

80 display blocks, 92 labels. Hashes bind a version, not formula semantics. Line numbers refer to paper/submitted.tex.

Label(s)	Line	SHA-256 prefix
eq:main-lower-bound	163	2ee11a326ea2f3
eq:boundaryconvention	378	26e2bccfd914f5
eq:normconvention	396	0558a08d702324
eq:BM	507	eed7e1157e86ec
eq:equiv	561	553859a25cafdb
eq:principal	588	93f6b485789023
eq:star	621	20c7cbbc7c5aae
eq:clutching	777	c2a8226dc99750
eq:connectionmetric	873	b798f3a36f72ac
eq:Lambda	890	c5b11126367f67
eq:smallradius	899	bf0f83042940b4
eq:brackets	940	b2a7620490743f
eq:verticalderivative	947	cc16a24c5f1410
eq:Koszul1, eq:Koszul2	951	db422dbfa07921
eq:HVHV, eq:HHVV, eq:HVVV, eq:HHHV	969	202e916117666f
eq:HHHH	1031	d1cae3c6f6a15d
eq:VVVV	1041	a8590e009ec3e4
eq:Plucker	1072	35cd0756bec03b
eq:bracketcontraction	1100	647094e960a3e2
eq:sharp-master	1153	ba8165ccfa6391
eq:master	1164	4bc69d7a010d85
eq:smallA, eq:smallB, eq:smallC	1191	a2569ef99c1973
eq:reserve	1227	693b9403e804cd
eq:southlowerbound	1233	830a64ad06e106

APPENDIX B / 25-48

Source formula inventory

80 display blocks, 92 labels. Hashes bind a version, not formula semantics. Line numbers refer to paper/submitted.tex.

Label(s)	Line	SHA-256 prefix
eq:south	1271	86355cc5f91342
eq:boundarydata	1282	fb2cd817de558d
eq:southsourceform	1296	325d5e83e61861
eq:potentials	1307	cabaed4001538f
eq:omega	1318	5a8358868933cf
eq:ONeill	1376	0b3304ada9ef28
eq:north	1416	f1c790b622f1e8
eq:northendpoints	1440	d95ce1564de684
eq:northsourceform	1448	04b16a585ab6eb
eq:horizontalgraph	1480	20702a00f9be71
eq:radialarea, eq:mixedarea	1489	7e4415e9348958
eq:warpedcurvature	1512	34aef055efbd76
eq:northcriterion	1543	1f954aaf1a8168
eq:delta	1617	db1ac041ae4b6f
eq:F	1622	0c9e01b7e8da88
eq:r	1633	e4228efb7db437
eq:epsnorth	1638	c90c44462659d3
eq:rproperties	1644	e21cf03ce9271d
eq:rjet	1651	c407383d848aa5
eq:density	1657	fd0f8dc0437e68
eq:Fcurvature	1666	33eb952a4a1cd7
eq:elementary	1682	44af1da546fd3a
eq:northlowerbound	1781	a00696d4b5029f
eq:boundarmargin	1829	3921102a6704b4

APPENDIX B / 49-72

Source formula inventory

80 display blocks, 92 labels. Hashes bind a version, not formula semantics. Line numbers refer to paper/submitted.tex.

Label(s)	Line	SHA-256 prefix
eq:sourceboundary	1840	1571d3a78ed3cd
eq:cometric	1854	267f093ec76dde
eq:forms	1880	5c0f963b02c361
eq:sum	1886	23347322e8eef4
eq:sum-eigenvalues	1901	c1925557460337
eq:principal-appendix	2049	0a246184acf55f
eq:glue-collars	2108	9a023a6eca191a
eq:glue-jump	2121	63b994b19ae23e
eq:glue-hermite	2139	78e8f60175cf41
eq:glue-endpoint-jets	2148	522135dbd87bc5
eq:glue-hermite-first, eq:glue-hermite-second	2153	928c5081fc98db
eq:glue-estimate-metric, eq:glue-estimate-first, eq:glue-estimate-second	2175	41a6bdc2433bff
eq:glue-christoffel	2211	79d4cc961035f0
eq:glue-gauss, eq:glue-radial, eq:glue-codazzi	2221	f742eac3aa5759
eq:glue-radial-sectional	2245	ddb61a72163872
eq:glue-tangent-sectional	2259	72d3b68abc43ab
eq:glue-determinant	2273	0edacfa3169393
eq:glue-tangent-bound	2284	47c53b14f6071f
eq:glue-tangent-reserve	2303	f8023f66bcf761
eq:glue-radial-reserve	2326	799a0c081ba34e
eq:glue-cross-reserve	2335	6b095bc0ff8b7b
eq:glue-all-planes	2356	3149261c6ce596
eq:glue-tensor-mollifier	2409	7214f760b3ff4e
eq:glue-C1-convergence	2427	a919c7628bb072

APPENDIX B / 73-80

Source formula inventory

80 display blocks, 92 labels. Hashes bind a version, not formula semantics. Line numbers refer to paper/submitted.tex.

Label(s)	Line	SHA-256 prefix
eq:glue-coordinate-curvature	2441	c3f69ee53920f2
eq:glue-linear-nonlinear	2450	f2fa584ade9990
eq:glue-curvature-error	2465	5cc52d5278e7be
eq:glue-error-control	2482	21f78134349e8a
eq:glue-metric-equivalence	2501	ca8ee2d01ee070
eq:glue-ae-positive	2509	b83a69912fd17f
eq:glue-smoothed-positive	2538	a6940402fd852d
eq:glue-localization	2551	c19e05ab2b0487